

SAT UZBEKISTAN

Telegram @satuzbhunters

Module-1

1  Mark for Review



On a particular assembly line, 51 out of every 100 products have a defect. A product from the assembly line will be selected at random. What is the probability of selecting a product that has a defect?

(A) $\frac{1}{5,100}$

(B) $\frac{1}{51}$

(C) $\frac{51}{100}$

(D) $\frac{51}{10}$

2



Mark for Review



The number of muskrat in a 131-acre area is estimated to be 2,096. What is the estimated population density, in muskrat per acre, of this area?

(A) 16

(B) 131

(C) 147

(D) 2,112

3



Mark for Review



Sabrina has a goal to save at least \$240. Sabrina has already saved \$190. If x represents the additional amount of money, in dollars, Sabrina needs to save to meet her goal, which inequality represents this situation?

(A) $\frac{x}{190} \geq 240$

(B) $190x \geq 240$

(C) $190 + x \geq 240$

(D) $190 - x \geq 240$

4



Mark for Review



The function f is defined by $f(x) = 10x^2 - 40x - 150$. What is the value of $f(0)$?

(A) -150

(B) -40

(C) 0

(D) 10

5

Mark for Review



$$y = 2x + 43$$

Which table gives three values of x and their corresponding values of y for the given equation?

A

x	y
0	0
1	45
2	47

B

x	y
0	2
1	45
2	47

Ⓒ

x	y
0	2
1	45
2	88

Ⓓ

x	y
0	43
1	45
2	47

@satuzbhunters

6

Mark for Review



In triangle ABC , $AB = 4,290$ millimeters (mm) and $BC = 4,290$ mm. Which statement is sufficient to prove that triangle ABC is equilateral?

- ☐ A $AC = 4.29$ mm
- ☐ B $AC = 42.9$ mm
- ☐ C $AC = 429$ mm
- ☐ D $AC = 4,290$ mm

7

Mark for Review



How many solutions does the equation $8x + 5 = 8x + 2$ have?

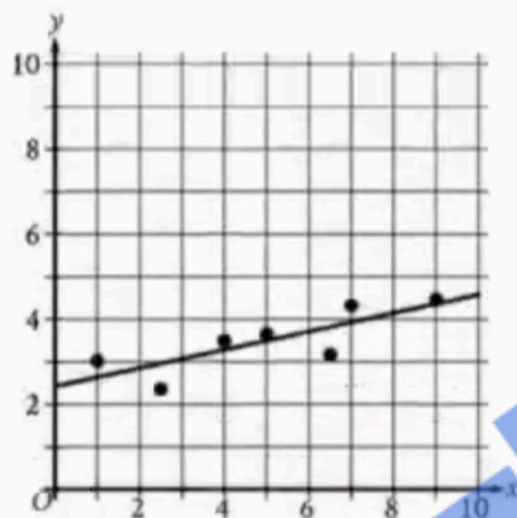
- ☐ A Zero
- ☐ B Exactly one
- ☐ C Exactly two
- ☐ D Infinitely many

8

Mark for Review

ABC

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of the line of best fit?

(A) -2.42

(B) -0.22

(C) 0.22

(D) 2.42

9



Mark for Review

$$y = 4x + 8$$

$$y = 3x + 1$$

In the xy -plane, the graphs of the given equations intersect at (x, y) . What is the value of x ?

Answer Preview:

10



Mark for Review

The equation $8p + 8w = 720$ represents the number of purple tiles, p , and the number of white tiles, w , a designer needs to cover a 720-square-inch section of a wall. The designer needs 34 purple tiles for the wall. How many white tiles does she need?

11

$$y = -6.5$$

$$y = x^2 + 8x + a$$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a ?

12



Mark for Review



Line t in the xy -plane has a slope of $\frac{1}{8}$ and passes through the point $(40, -7)$. Which equation defines line t ?

(A) $y = -12x + \frac{1}{8}$

(B) $y = \frac{x}{8} - 12$

(C) $y = \frac{x}{8} - 7$

(D) $y = 40x - 7$

13



Mark for Review



A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 184(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 184-gram sample starts to decay. Which statement is the best interpretation of $s(87) = 23$ in this context?

- (A) Approximately 23 grams of the sample remains 87 years after the sample starts to decay.
- (B) The mass of the sample has decreased by approximately 23 grams 87 years after the sample starts to decay.
- (C) The mass of the sample has decreased by approximately 87 grams 23 years after the sample starts to decay.
- (D) Approximately 87 grams of the sample remains 23 years after the sample starts to decay.

14



Mark for Review



$$ak = \frac{b}{10}(9 + m)$$

The given equation relates the positive variables a , b , k , and m . Which equation correctly expresses m in terms of a , b , and k ?

(A) $m = \frac{10(ak-9)}{b}$

(B) $m = ak - 9 - \frac{b}{10}$

(C) $m = ak - \frac{9b}{10}$

(D) $m = \frac{10ak}{b} - 9$



An object is launched from a height of 130 feet above the ground. A quadratic function f models the height above the ground, in feet, of the object t seconds after it is launched. According to the model, 2 seconds after the object is launched, it reaches a maximum height of 194 feet above the ground. Which equation defines f ?

(A) $f(t) = -16(t + 194)^2 - 2$

(B) $f(t) = -16(t - 194)^2 + 2$

(C) $f(t) = -16(t + 2)^2 - 194$

(D) $f(t) = -16(t - 2)^2 + 194$

16



Mark for Review



$$x + 4y = 2$$

$$x - 4y = 8$$

The solution to the given system of equations is (x, y) . What is the value of x ?

(A) 1.25

(B) 2

(C) 5

(D) 10

17

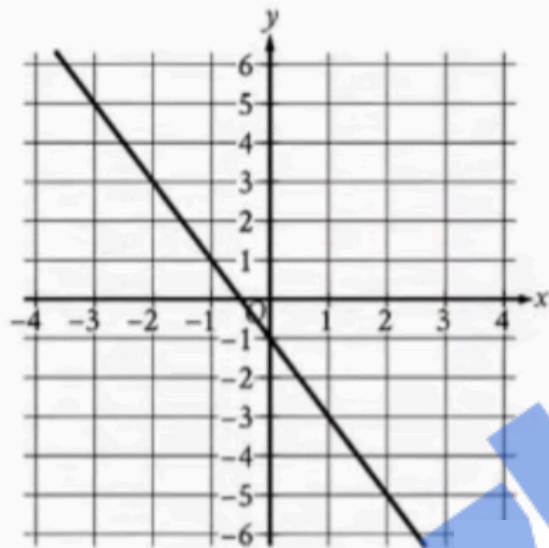


Mark for Review

In the xy -plane, the graph of the equation $(x - 6)^2 + (y - 10)^2 = 36$ is a circle. The point $(12, c)$, where c is a constant, lies on this circle. What is the value of c ?

Answer Preview:

The graph of $y = f(x) - 9$ is shown.



Which equation defines the linear function f ?

(A) $f(x) = -11x + 8$

(B) $f(x) = -11x - 9$

(C) $f(x) = -2x - 9$

(D) $f(x) = -2x + 8$

19



Mark for Review



A right square prism has a height of 16 units. The volume of the prism is 64 cubic units. What is the length, in units, of an edge of the base?

(A) 2

(B) 4

(C) 8

(D) 16



In triangle ABC , the measure of angle C is 90 degrees. The cosine of angle A is $\frac{60}{61}$. What is the sine of angle B ?

(A) $\frac{11}{61}$

(B) $\frac{11}{60}$

(C) $\frac{60}{61}$

(D) $\frac{61}{60}$



$$f(x) = 250(5)^x$$

The function f is defined by the given equation. If $g(x) = f(x - 2)$, which of the following equations defines the function g ?

(A) $g(x) = 10(5)^x$

(B) $g(x) = 125(5)^x$

(C) $g(x) = 250(25)^x$

(D) $g(x) = 250(10)^x$

$$ax^2 + 176x + c$$

In the given expression, a and c are positive constants. If $px + q$ is a factor of the expression, where p and q are positive constants, what is the greatest possible value of ac ?

SAT UZBEKISTAN

Telegram @satuzbhunters

Module-2

1



Mark for Review



The function f is defined by $f(x) = 35x + 60$. What is the value of $f(x)$ when $x = 2$?

(A) 70

(B) 97

(C) 130

(D) 190

2



Mark for Review



Which expression is equivalent to $90y + 80y$?

(A) $720y^2$

(B) $720y$

(C) $170y^2$

(D) $170y$

3



Mark for Review



Task	Time (minutes)
A	3
B	2
C	14
D	8
E	8

The table shows the amount of time it took a participant in a study to complete each of 5 tasks. What was the mean time, in minutes, for this participant to complete a task?

(A) 7

(B) 8

(C) 12

(D) 14

4



Mark for Review



If $6(x + 1) = 36$, what is the value of $x + 1$?

(A) 5

(B) 6

(C) 29

(D) 30

5



Mark for Review



The graph of the polynomial function f in the xy -plane, where $y = f(x)$, has x -intercepts of $(-5, 0)$ and $(2, 0)$. Which of the following must be true?

(A) $f(0) = -5$

(B) $f(2) = -5$

(C) $f(-5) = 0$

(D) $f(-5) = 2$

6

Mark for Review



$$y > 9x - 6$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

(A)

x	y
3	14
5	32
8	59

(B)

x	y
3	21
5	39
8	66

(C)

x	y
3	28
5	32
8	66

(D)

x	y
3	28
5	46
8	73

7



Mark for Review



When the linear function h is graphed in the xy -plane, where $y = h(x)$, its slope is -6 . If $h(2) = 1$, which equation defines h ?

(A) $h(x) = -6x + 13$

(B) $h(x) = -6x + 1$

(C) $h(x) = -6x + 7$

(D) $h(x) = -6x - 11$

8



Mark for Review

A linear model estimates the population of a city from 1995 to 2019. The model estimates the population was 56 thousand in 1995, 215 thousand in 2015, and x thousand in 2019. To the nearest whole number, what is the value of x ?

9



Mark for Review



For $x > 0$, the function f is defined as follows:

$f(x)$ equals 293% of x

Which of the following could describe this function?

(A) Decreasing exponential

(B) Decreasing linear

(C) Increasing exponential

(D) Increasing linear

10



Mark for Review



$$\frac{7}{2}x + 5y = 32$$

$$\frac{3}{2}x + 5y = 28$$

The solution to the given system of equations is (x, y) . What is the value of $\frac{47}{2}x + 15y$?

(A) 2

(B) 5

(C) 60

(D) 88

11



Mark for Review



$$\frac{1}{x} + \frac{1}{x-9} = \frac{4}{x^2-9x}$$

What are all the solutions to the given equation?

(A) -1 and 4

(B) 0 and 9

(C) 2

(D) $\frac{13}{2}$

12



Mark for Review

According to a set of standards, pure biodiesel fuel can contain a maximum of 0.001% phosphorus by mass. If a sample of pure biodiesel fuel has a mass of 290 grams, what is the maximum mass, in grams, of phosphorus the sample can contain to meet these standards?

13



Mark for Review

A proposal for a construction project was included on a city election ballot. A TV news report stated that 8 times as many people voted in favor of the proposal as people who voted against it. An internet article reported that 66,500 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

14



Mark for Review



$$x^2 - 12x + c = 0$$

In the given equation, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n ?

(A) -36

(B) -12

(C) 12

(D) 36

15



Mark for Review

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(2, 0)$ and a y -intercept at $(0, -15)$. What is the value of b ?

16



Mark for Review

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 12$ centimeters, and the height of the triangle is $x - 2$ centimeters. What is the value of x ?

17



Mark for Review



$$y = -4x^2 - 47$$

$$y = qx - 43$$

In the given system of equations, q is a positive constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

(A) -90

(B) -51

(C) -8

(D) -1

18



Mark for Review

$$56x^2 + (56s + r)x + rs = 0$$

In the given equation, r and s are positive constants. The product of the solutions to the given equation is krs , where k is a constant. What is the value of k ?

19



Mark for Review



In December 2017, the lowest temperature recorded in a certain city was 10 degrees Fahrenheit ($^{\circ}\text{F}$) and the highest temperature recorded was 90 $^{\circ}\text{F}$. Which inequality is true for all values of t , where t represents any temperature, in $^{\circ}\text{F}$, recorded in the city in December 2017?

(A) $10 \leq t \leq 90$

(B) $t \leq 10$

(C) $t \leq 80$

(D) $t \geq 90$



The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	1,440	480
Rectangle B	5,760	n

What is the value of n ?

(A) 960

(B) 1,920

(C) 3,840

(D) 4,800

21



Mark for Review



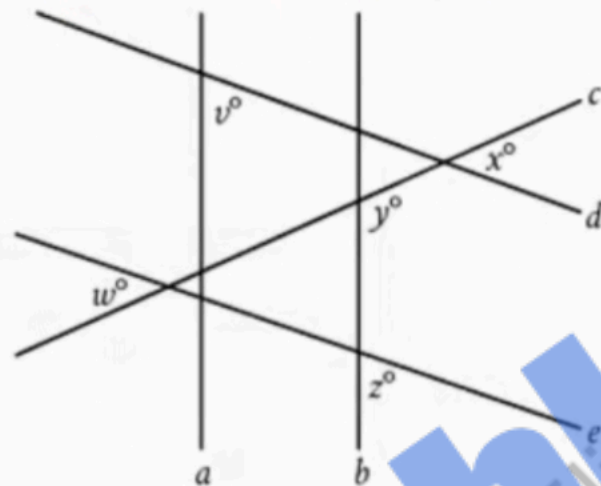
The cost of renting a piece of construction equipment for up to 10 days is \$370 for the first day and \$185 for each additional day. Which of the following equations gives the cost y , in dollars, of renting the equipment for x days, where x is a positive integer and $x \leq 10$?

(A) $y = 370x - 185$

(B) $y = 185x + 185$

(C) $y = 185x + 370$

(D) $y = 370x + 185$



Note: Figure not drawn to scale.

In the figure, parallel lines a and b are intersected by lines c , d , and e . If $z = 77$, $y = 108$, and $v < z$, which statement about x and w must be true?

- (A) $x < w$
- (B) $x > w$
- (C) $x = w$
- (D) $x + w = 90$